

Data warehouse Introduction

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Decision support systems

Decision support systems are

- Huge operational databases  available in most of the companies
 - these databases may provide a large wealth of useful information
- Decision support systems provide means for
 - in depth analysis of a company's business
 - faster and better decisions

Strategic decision support

- Demand evolution analysis and forecast
- Critical business areas identification
- Budgeting and management transparency
 - reporting, practices against frauds and money laundering
- Identification and implementation of winning strategies
 - cost reduction and profit increase

Business Intelligence

- BI provides support to strategic decision support in companies
- Objective: ^{Our main goal is} transforming company data into actionable information
 - at different detail levels
 - for analysis applications
- Users may have heterogeneous needs
- BI requires an appropriate hardware and software infrastructure

Applications

- Manufacturing companies: order management, client support
- Distribution: user profile, stock management
- Financial services: buyer behavior (credit cards)
- Insurance: claim analysis, fraud detection
- Telecommunication: call analysis, churning, fraud detection
- Public service: usage analysis
- Health: service analysis and evaluation

Data warehouse

A data warehouse is a

- Database devoted to decision support, which is kept separate from company operational databases
- They are:
 - devoted to a specific subject
 - Integrated and consistent
 - time dependent, non volatile

used for decision support in a company

W. H. Inmon, Building the data warehouse, 1992

Why separate data?

- Performance
 - complex queries reduce performance of operational transaction management
 - different access methods at the physical level
- Data management
 - missing information (e.g., history)
 - data consolidation
 - data quality (inconsistency problems)

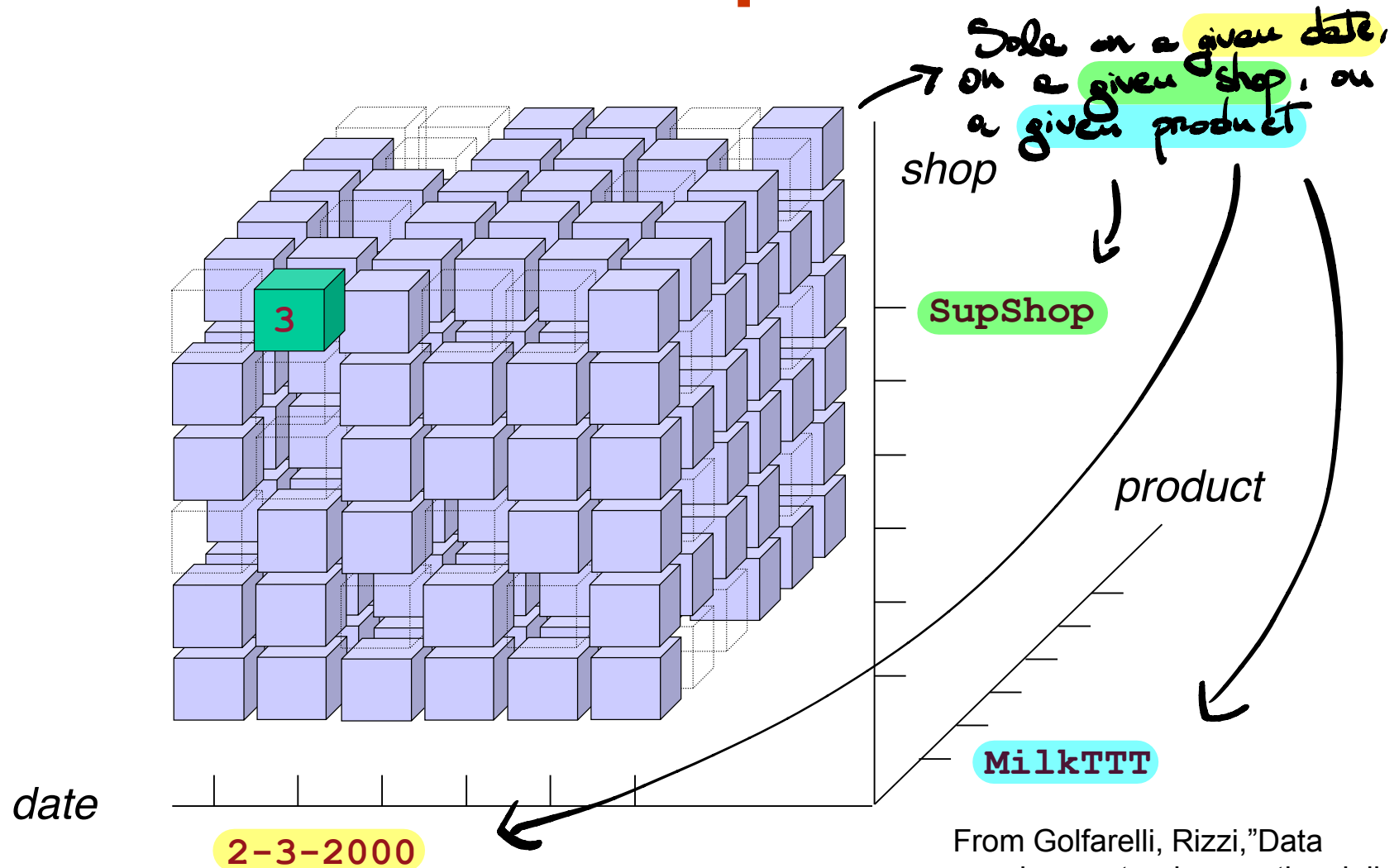
Data structure and data analysis

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Multidimensional representation

- Data are represented as an (hyper)cube with three or more dimensions
- Measures on which analysis is performed: cells at dimension intersection
- Data warehouse for tracking sales in a supermarket chain:
 - dimensions: product, shop, time
 - measures: sold quantity, sold amount, ...

Multidimensional representation



From Golfarelli, Rizzi, "Data warehouse, teoria e pratica della progettazione", McGraw Hill 2006

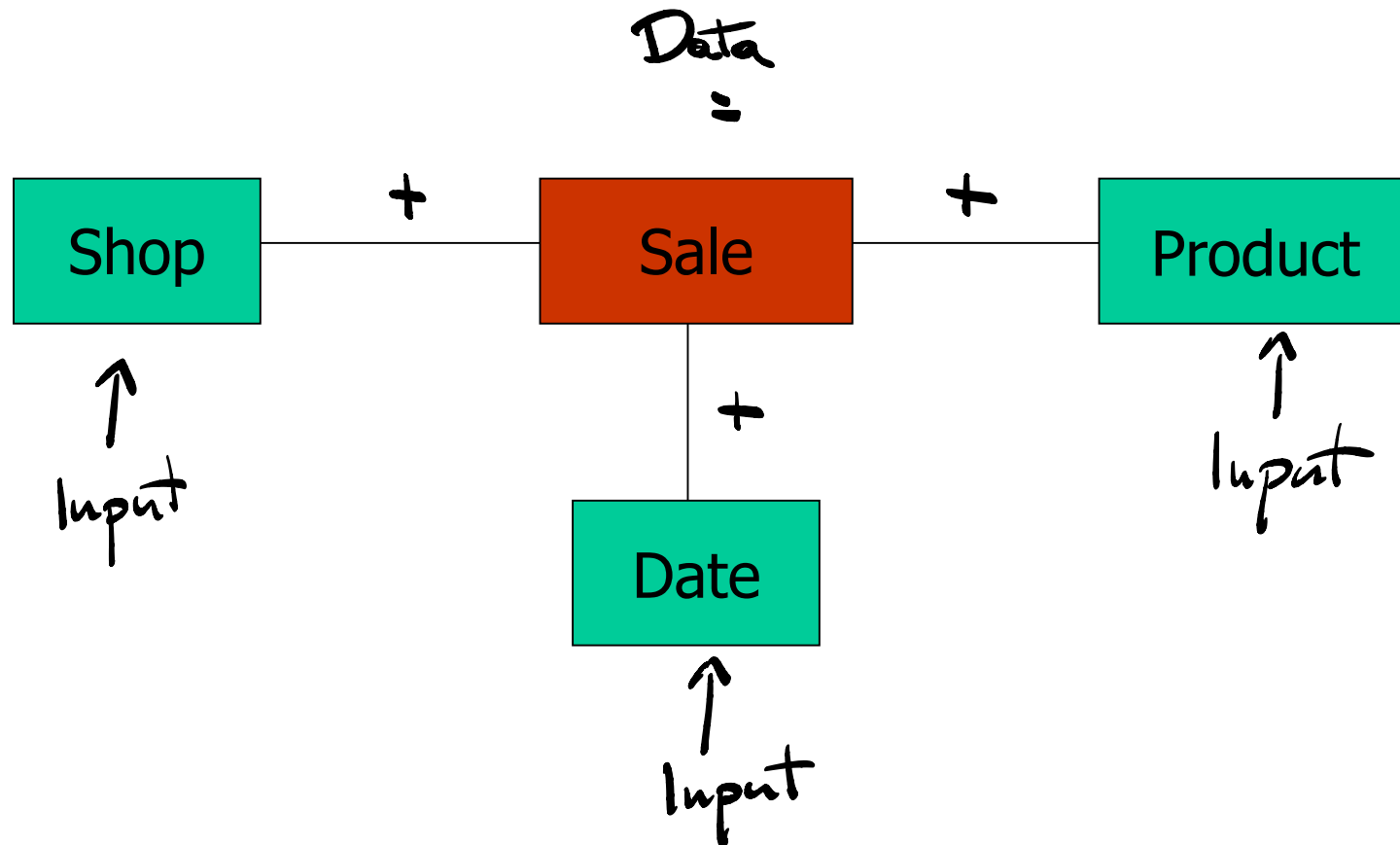
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Relational representation: star model

- Numerical measures stored in the *fact table*
 - attribute domain is numeric
- *Dimensions* describe the context of each measure in the fact table
 - characterized by many descriptive attributes

Example

Data warehouse for tracking sales in a supermarket chain



Data warehouse size

For instance, referring to a shop, we have the following dimensions:

- 1 • Time dimension: 2 years x 365 days
 - 2 • Shop dimension: 300 shops
 - 3 • Product dimension: 30.000 products, of which 3.000 sold every day in every shop
- Number of rows in the fact table:

$$730 \times 300 \times 3000 = 657 \text{ millions}$$

⇒ Size of the fact table $\approx$ 21GB

Data analysis tools

We have different kinds of analysis available, such as:

- **OLAP analysis:** complex aggregate function computation
 - support to different types of aggregate functions (e.g., moving average, top ten)
- **Data analysis by means of data mining techniques**
 - various analysis types
 - significant algorithmic contribution

↳ machine learning

Data analysis tools

- Presentation
 - separate activity: data returned by a query may be rendered by means of different presentation tools
- Motivation search
 - Data exploration by means of progressive, “incremental” refinements (e.g., drill down)

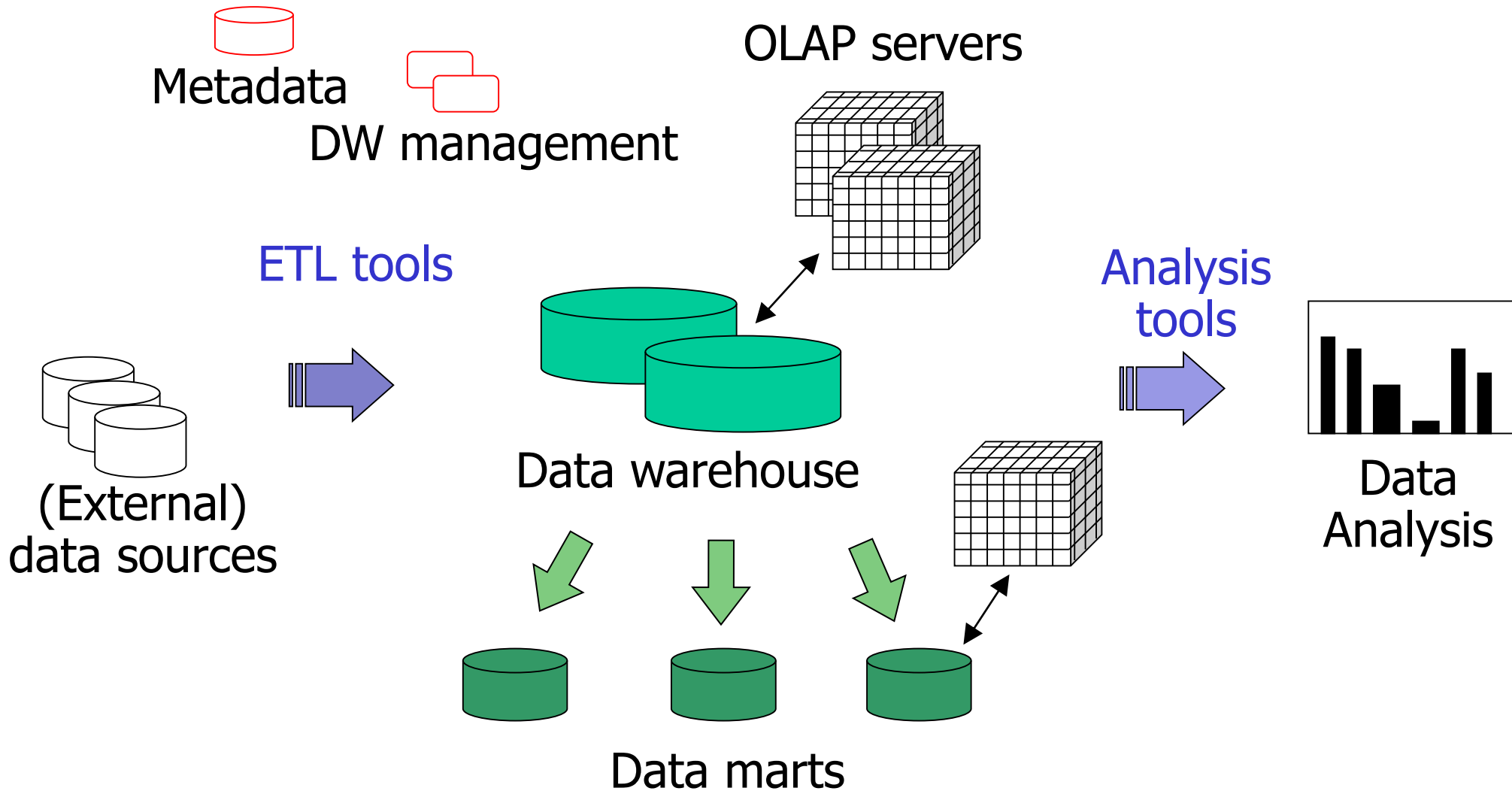
Data warehouse architectures

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Data warehouse architectures

- Separation between transactional computing and data analysis
 - avoid one level architectures
- Architectures characterized by two or more levels
 - separate to a different extent data incoming into the data warehouse from analyzed data
 - more scalable

Data warehouse: architecture



Data warehouse and data mart

A **Company data warehouse:** ☐ contains *all* the information on the company business and can be considered a collection of data marts

- extensive functional modelling process
- design and implementation require a long time

A **Data mart:** ^{is a} departmental information subset focused on a given subject $\hookrightarrow$ A portion of the data warehouse focused on a specific subject

- two architectures
 - dependent, fed by the company data warehouse
 - independent, fed directly by the sources
- faster implementation
- requires careful design, to avoid subsequent data mart integration problems

Servers for Data Warehouses

We have different kinds of servers for data warehouses:

- ROLAP (Relational OLAP) server
 - which is an*
 - extended relational DBMS
 - compact representation for sparse data
 - SQL extensions for aggregate computation
 - with*
 - specialized access methods which implement efficient OLAP data access
- MOLAP (Multidimensional OLAP) server
 - within which*
 - data represented in proprietary (multidimensional) matrix *one* format
 - sparse data require compression
 - special OLAP primitives
- HOLAP (Hybrid OLAP) server

Extraction, Transformation and Loading (ETL)

The ETL phase

- Prepares data to be loaded into the data warehouse and is divided into 4 further phases:
 - data extraction from (OLTP and external) sources
 - data cleaning
 - data transformation
 - data loading
- It is* performed
 - when the DW is first loaded
 - during periodical DW refresh

ETL process

- *Data extraction*: data acquisition from sources
- *Data cleaning*: techniques for improving data quality (correctness and consistency)
- *Data transformation*: data conversion from operational format to data warehouse format
- *Data loading*: update propagation to the data warehouse

Metadata

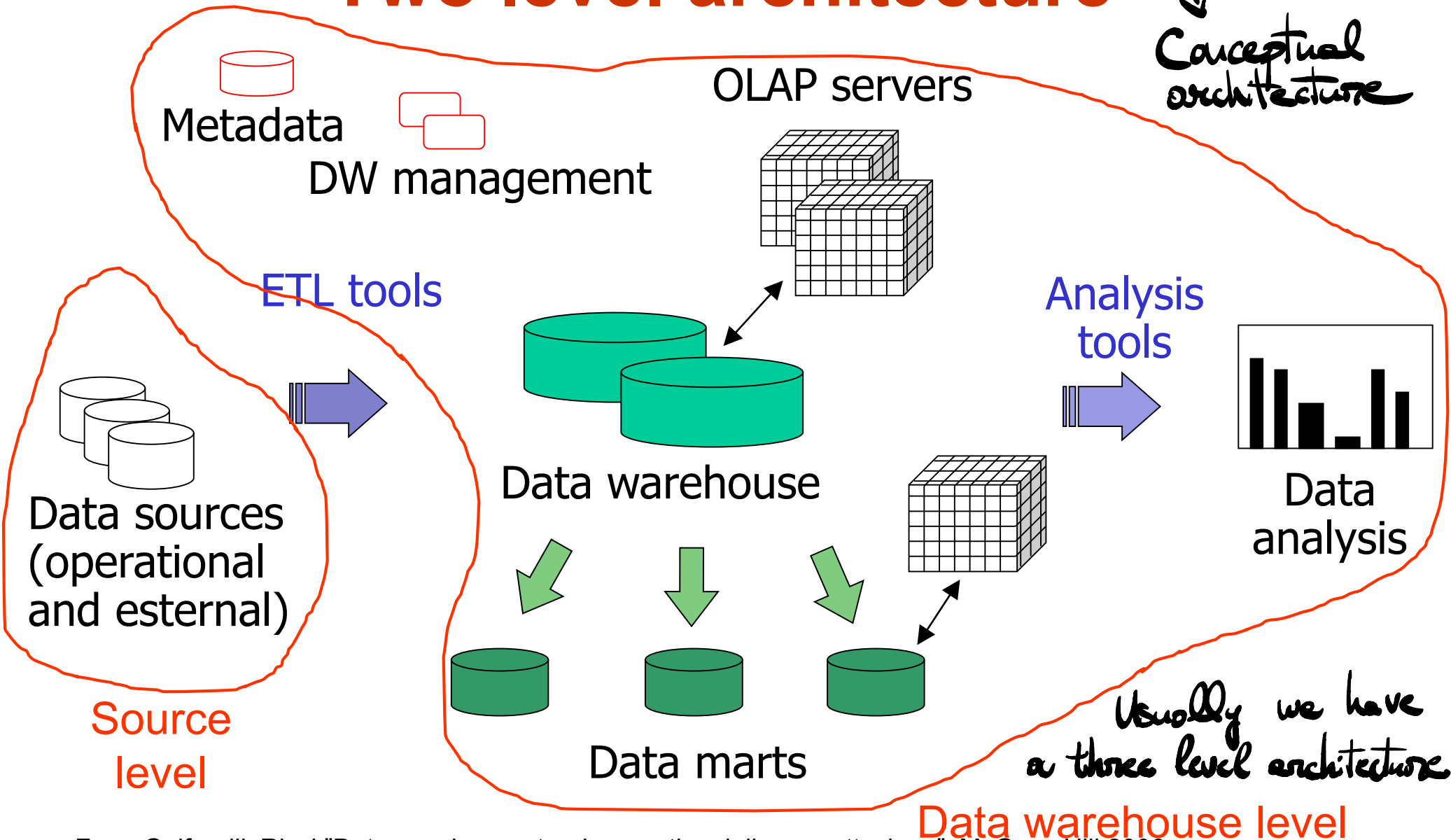
→ information which describe data

metadata = data about data

- Different types of metadata:
 - for data transformation and loading: describe data sources and needed transformation operations
 - Useful using a common notation to represent data sources and data after transformation
 - CWM (Common Warehouse Metadata Initiative): standard proposed by OMG to exchange data between DW tools and repository of metadata in heterogeneous and distributed environments
 - for data management: describe the structure of the data in the data warehouse
 - also for materialized view
 - for query management: data on query structure and to monitor query execution
 - SQL code for the query
 - execution plan
 - memory and CPU usage

Two level architecture

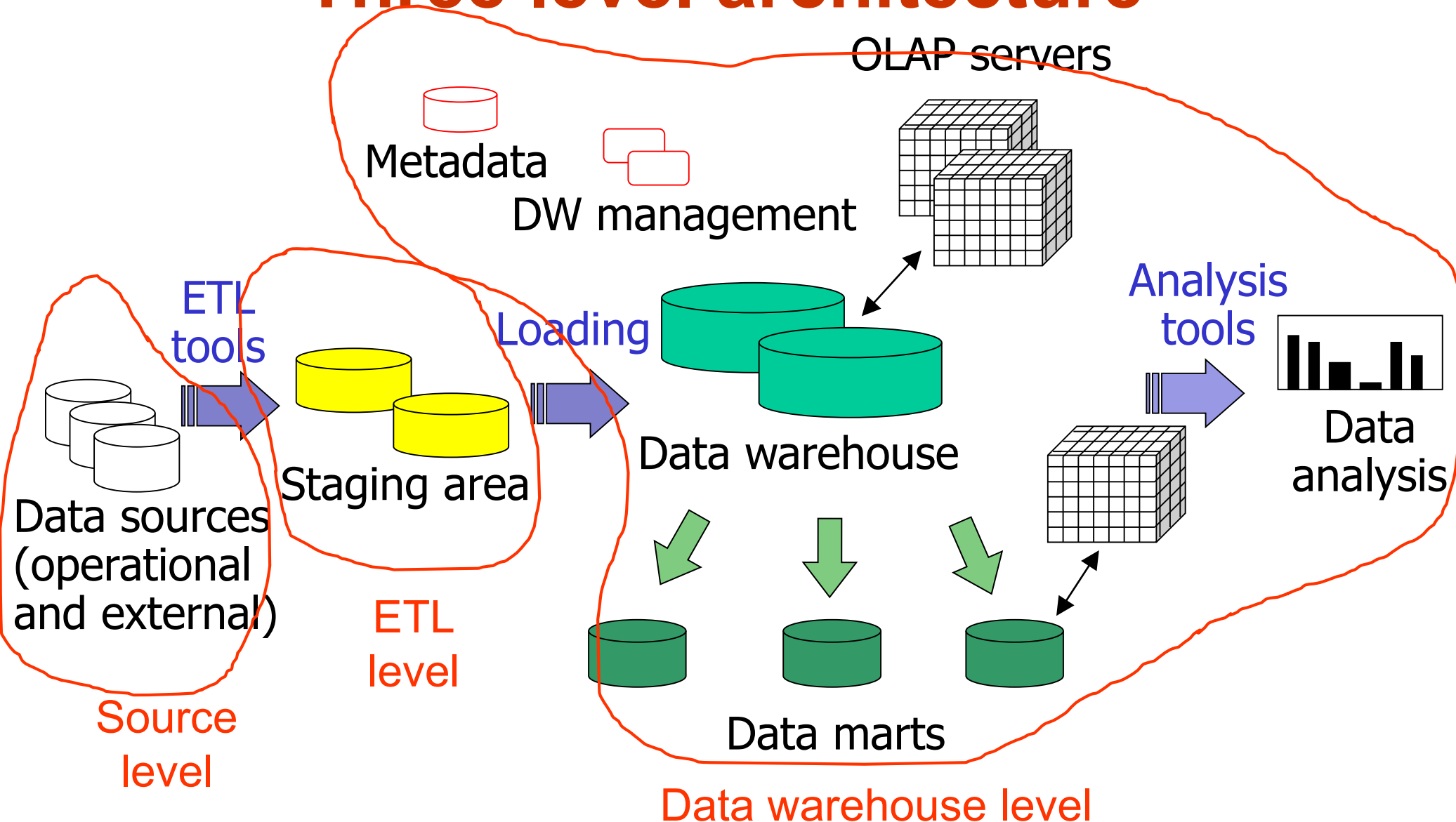
Conceptual architecture



Two level architecture features

- Decoupling between source and DW data
 - management of external (not OLTP) data sources (e.g., text files)
 - data modelling suited for OLAP analysis
 - physical design tailored for OLAP load
- Easy management of different temporal granularity of operational and analytical data
- Partitioning between transactional and analytical load
- “On the fly” data transformation and cleaning (ETL)

Three level architecture



From Golfarelli, Rizzi, "Data warehouse, teoria e pratica della progettazione", McGraw Hill 2006

Three level architecture features

The is a

- **Staging area:** buffer area allowing the separation between ET management and data warehouse loading → better reliability paying a greater overhead
 - complex transformation and cleaning operations are eased
 - provides an integrated model of business data, still close to OLTP representation
 - sometime denoted as Operational Data Store (ODS)
- **Introduces further redundancy**
 - more disk space is required for data storage